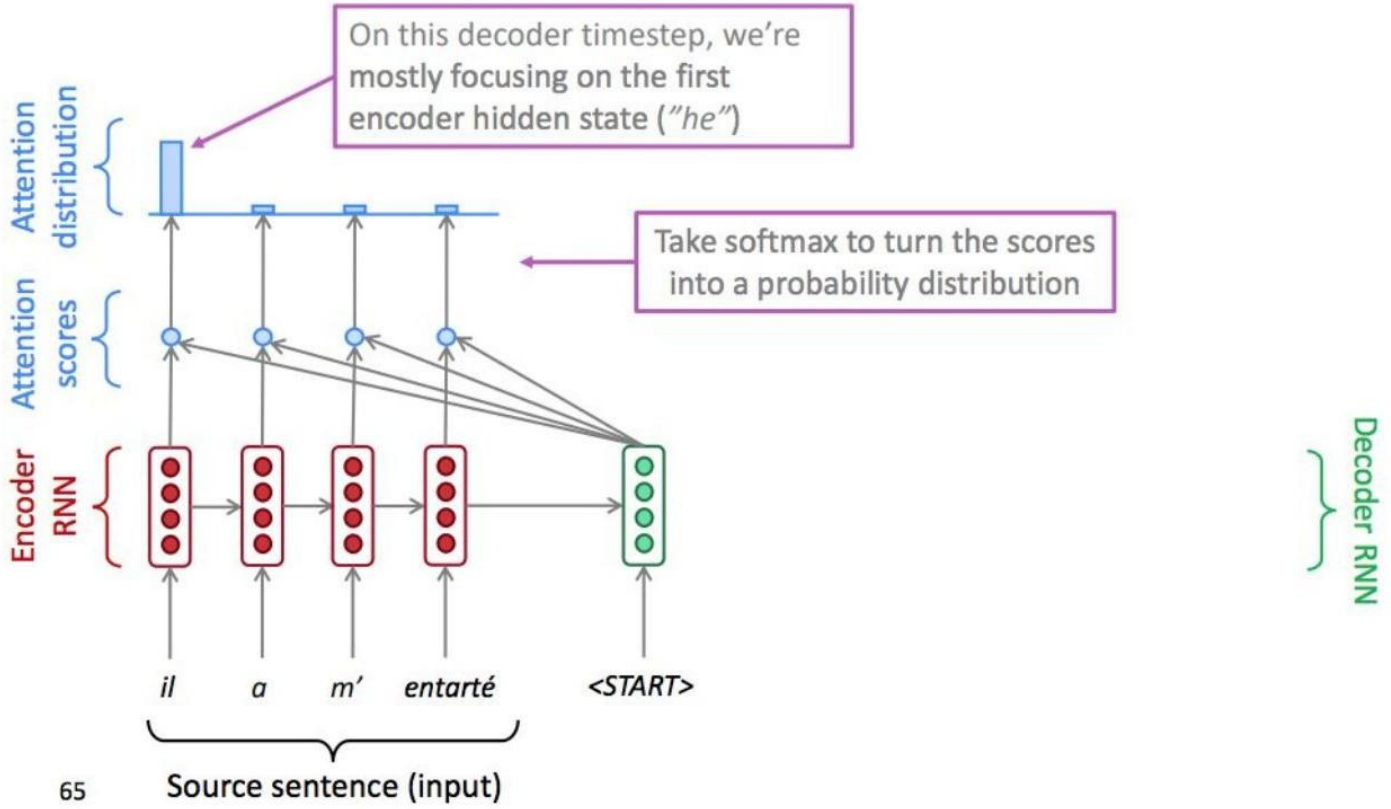
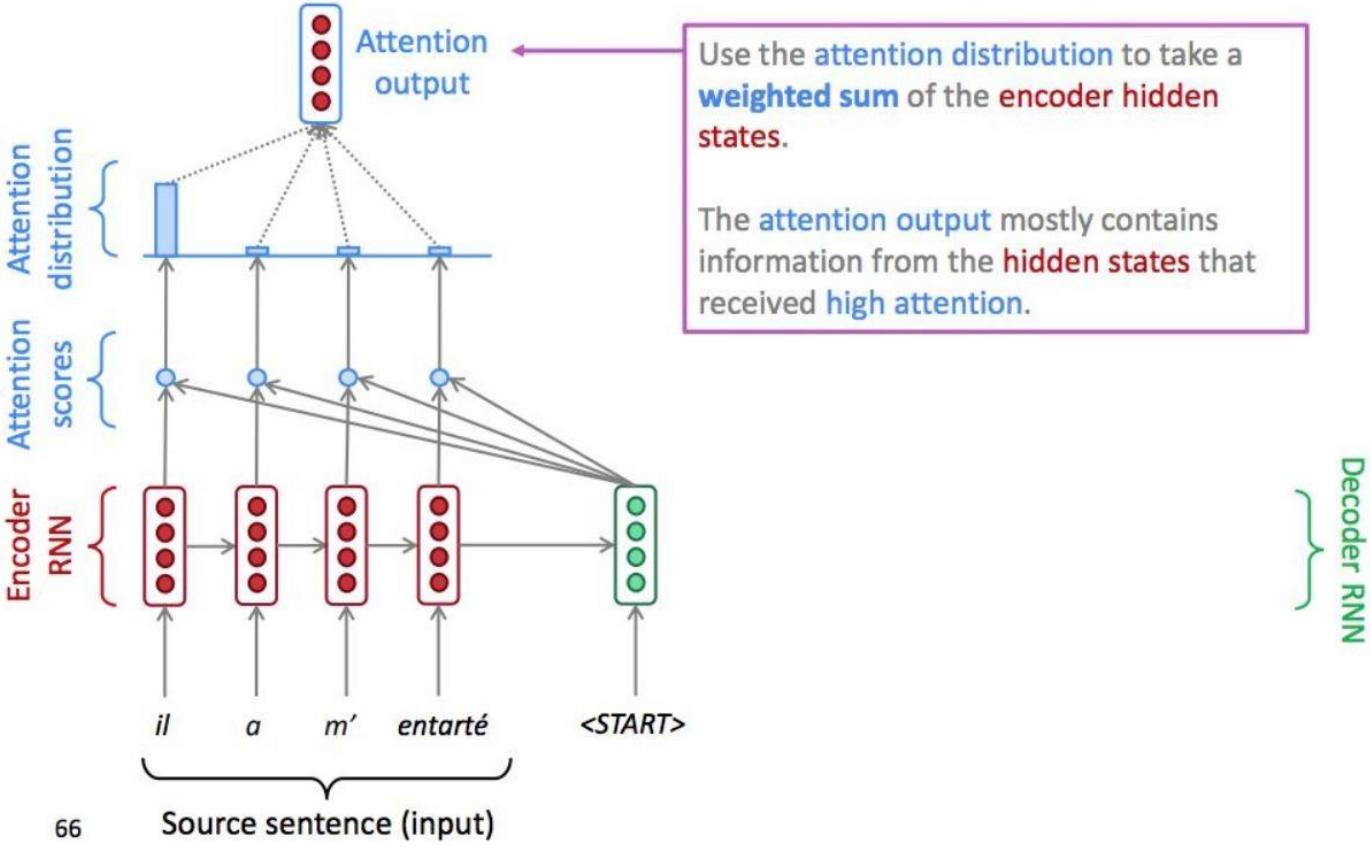
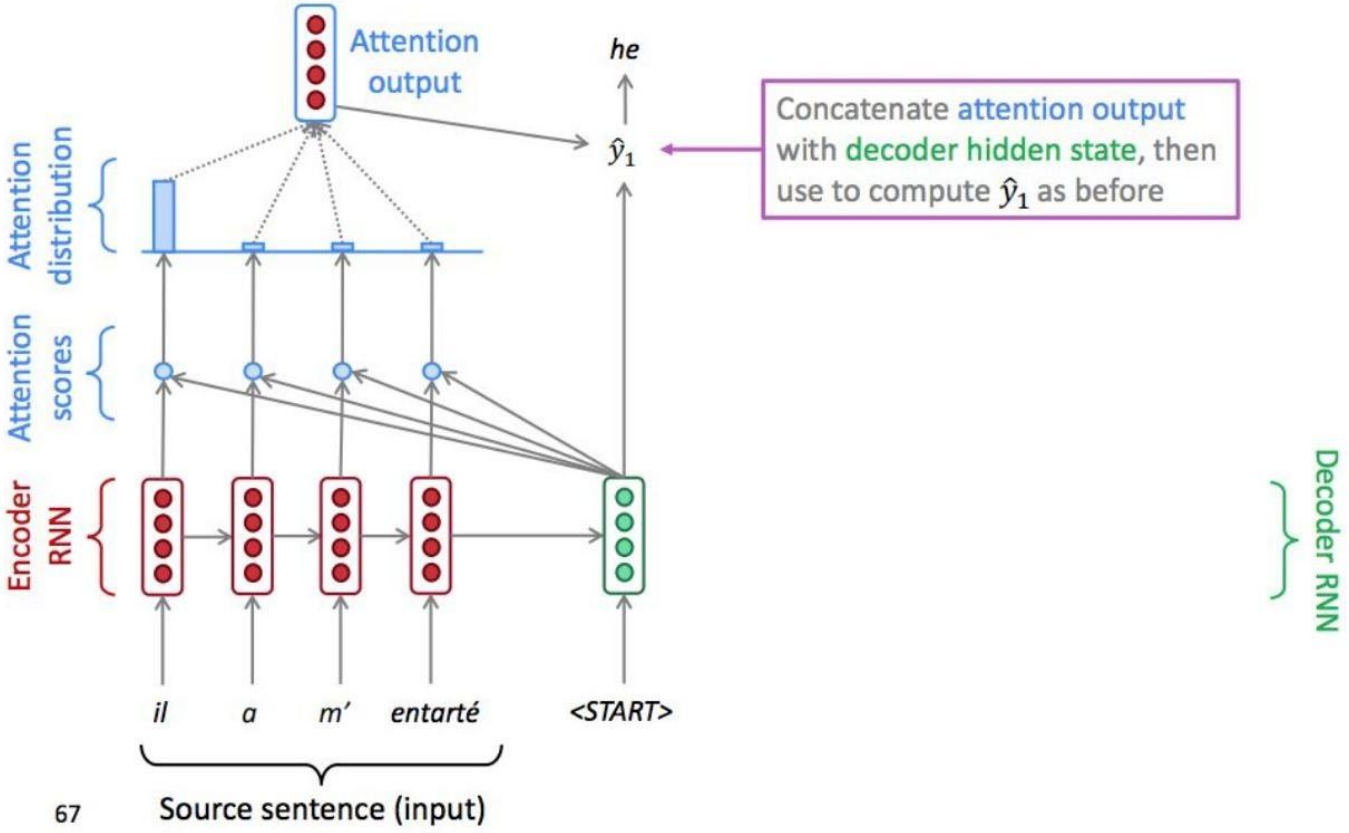
RNN - Attention



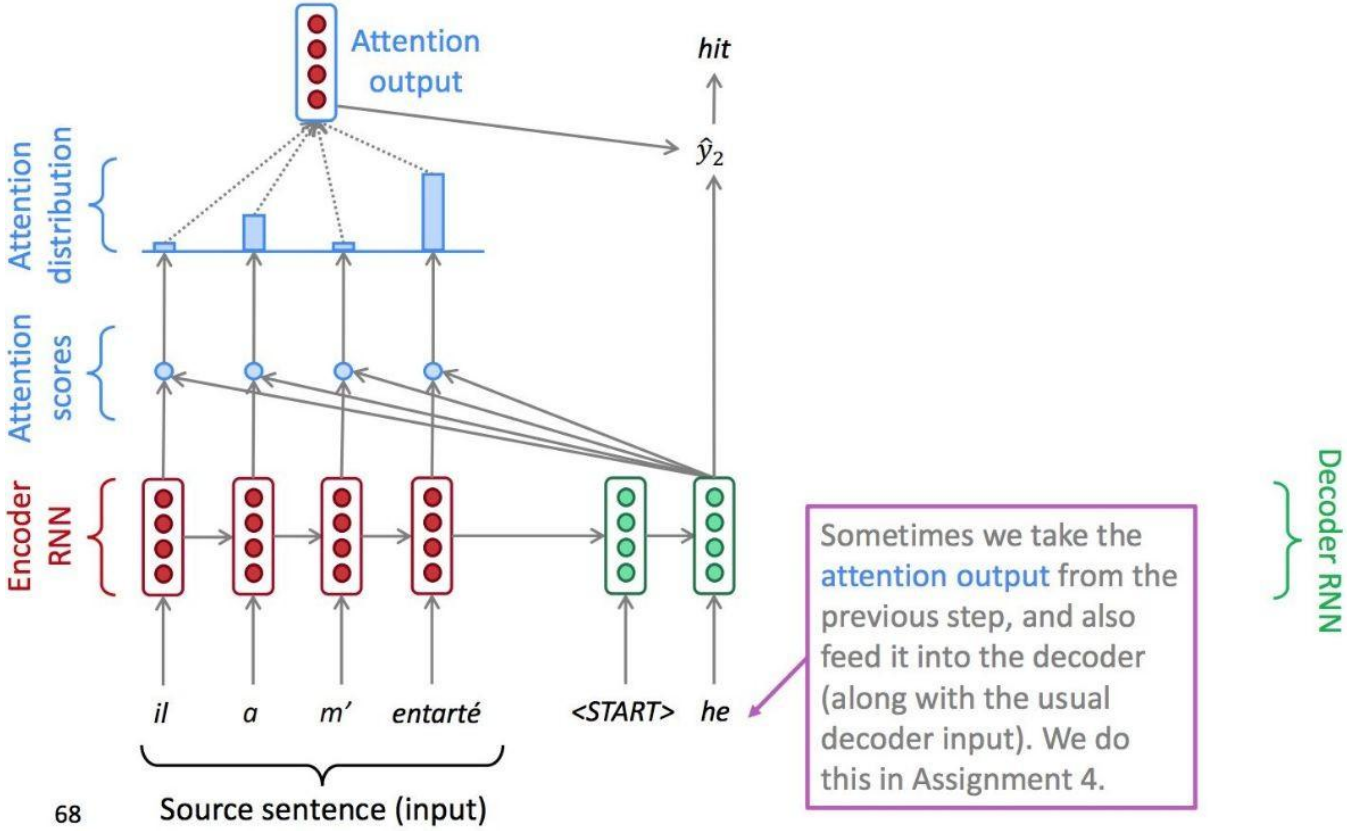
RNN - Attention



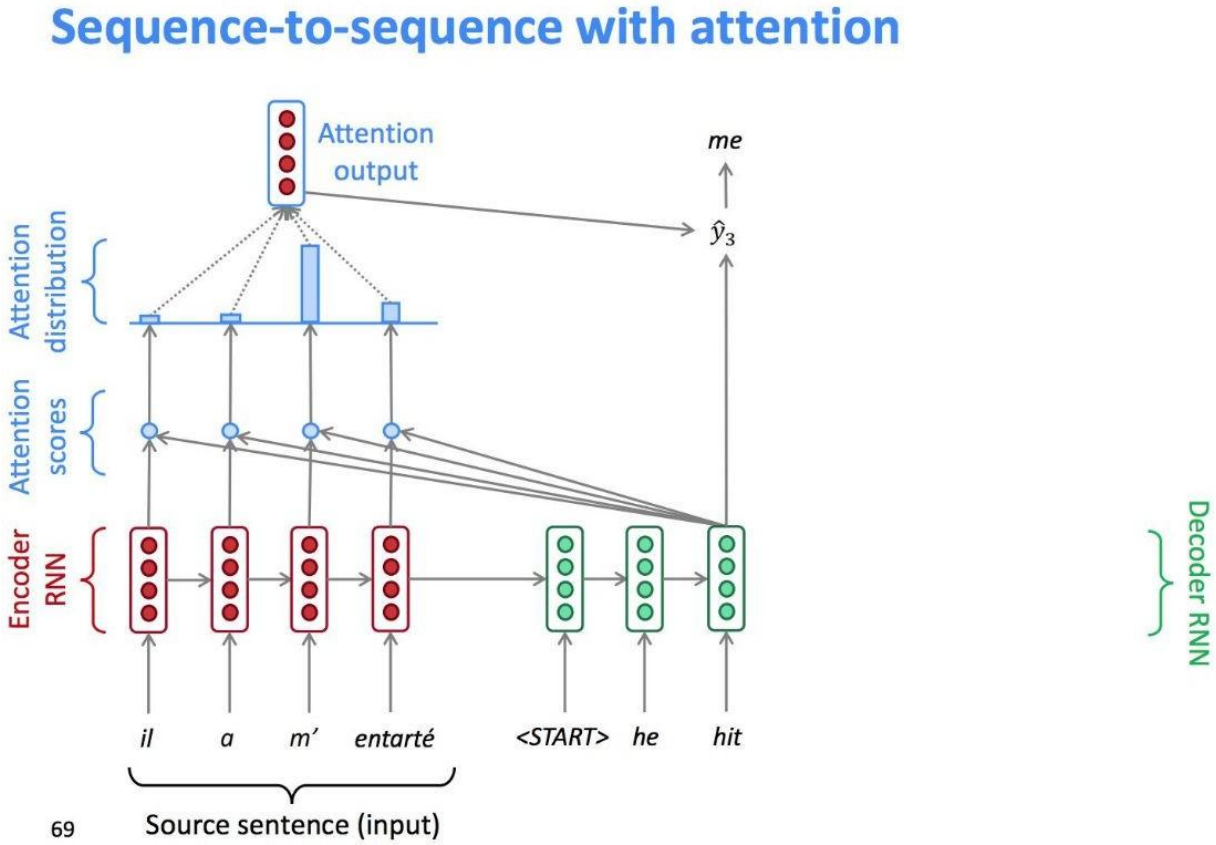
RNN - Attention



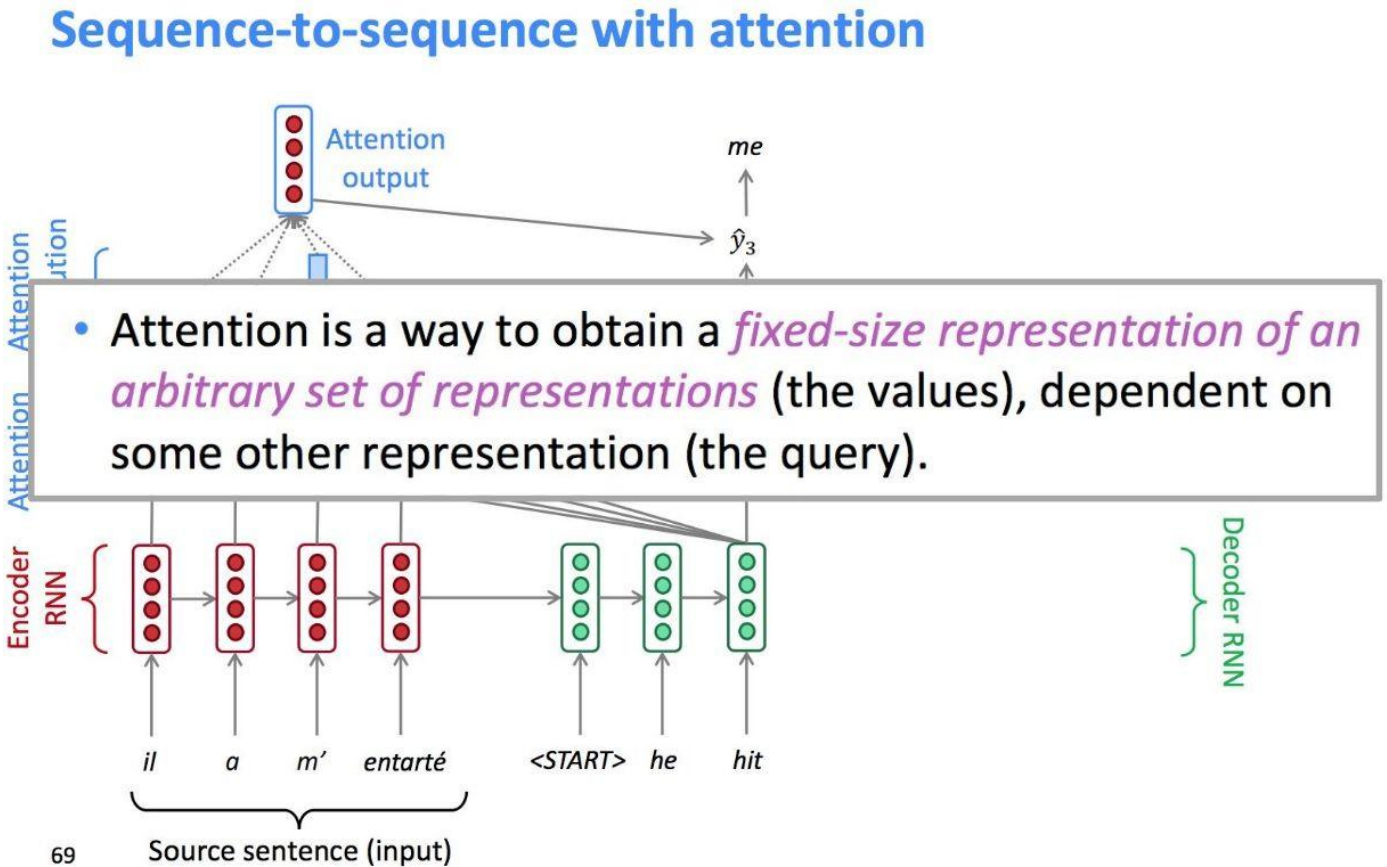
RNN - Attention



RNN - Attention



RNN - Attention



Attention

- For query vector **q**, key vector $\mathbf{k}_i$ representing value $\mathbf{v}_i$
  - s_i is the similarity score between **q** and $\mathbf{k}_i$
- Normalize the similarity scores to sum to 1
  - $p_i = \text{Softmax}(s_i)$
- Compute **z** as the weighted sum of the value vectors $\mathbf{v}_i$ weighted by their scores p_i
- In Machine Translation & Image Captioning, the keys and values are the same.
  - But, they could be different.
-

$$z = \sum_{i=1}^{L} p_i v_i$$

Attention is great

- Attention significantly improves performance (in many applications)
  - It's very useful to allow decoder to focus on certain parts of the source
- Attention solves the bottleneck problem
  - Attention allows decoder to look directly at source; bypass bottleneck
- Attention helps with vanishing gradient problem
  - Provides shortcut to faraway states
- Attention provides some interpretability
  - By inspecting attention distribution, we can see what the decoder was focusing on